Sameer Khan

SENIOR MACHINE LEARNING ENGINEER | DOCUMENT AI & MULTIMODAL AI | LLM/RAG | AWS MLOps

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Machine Learning Engineering professional targeting **Senior/Lead roles**, with extensive experience in building scalable, production-grade **AI/ML systems** and enterprise solutions. Expertise across **Machine Learning, Deep Learning, NLP, LLMs, ML pipelines, and deployment**, with an architecture-oriented approach to delivering **high-performance solutions** as a senior technical individual contributor.

PROFILE SUMMARY

- **Senior Machine Learning Engineer with 9+ years of experience** in building and productionizing **enterprise-scale AI/ML solutions**, with end-to-end exposure across **model development, evaluation, optimization, deployment,** and **production operations**.
- Proven experience in delivering **large-scale intelligent automation solutions**, including a **643-class mortgage solution processing 1.7M requests/day with 99.99% success** and production processing of **2.8M+ pages across 49 encrypted lender sources**.
- Strong background in **Deep Learning, Computer Vision, NLP, Multimodal Learning, and Generative AI**, with experience developing and optimizing models for complex enterprise use cases and achieving up to **0.975 weighted F1** through distributed and multi-GPU training.
- Practical experience in building **LLM and RAG-based applications**, including retrieval, reranking, structured generation, and model adaptation, with exposure to **modern GenAI and Agentic AI architectures**.
- Experienced in engineering **scalable ML pipelines and production platforms**, with a focus on **performance optimization, deployment automation, monitoring, reliability, and operational efficiency**, taking solutions from experimentation through production.
- **Architecture-oriented technical individual contributor** experienced in contributing to **ML system design, production reliability, incident RCA, cost optimization, and release governance**, targeting **Senior/Lead Machine Learning Engineering roles** focused on solving complex, large-scale ML challenges.

WORK EXPERIENCE

ICE DATA SERVICES | Senior ML Engineer | Hyderabad | Nov 2023 – Present

Highlights:

- **Enterprise ML Scale:** Supported a mortgage document classification platform processing **1.7M requests/day and 45M+ requests/month** with **99.99% success rate** and **1.8-second p95 latency**.
- **Business Transformation:** Reduced document verification turnaround from **~4 days to 4–5 hours** through production ML automation.
- **Model Excellence:** Delivered **0.975 weighted F1** on **528K+ pages**, outperforming the existing OCR-based production ensemble.
- **Accuracy Recovery:** Restored production F1 from **0.62 to 0.92** across **946,355 pages** by resolving a train-to-benchmark taxonomy mismatch and improving prediction calibration.
- **Cost Optimization:** Achieved approximately **20% lower hourly infrastructure cost** through model and deployment optimization.
- **Production Leadership:** Delivered **4 consecutive ML releases** and scaled production infrastructure from **20 to 118 SageMaker instances** while maintaining platform reliability.

Key Result Areas:

- **Own the end-to-end ML lifecycle** across model development, evaluation, optimization, deployment, monitoring, and production operations for **643 document classes and 49 encrypted lender sources**.
- **Develop and scale multimodal classification architectures** using **ResNet50, CNN, BiLSTM, and custom multi-head-attention Transformers**; distribute training across **20 A10G GPUs using Horovod**, resolving host-memory OOM and distributed all-reduce failures and achieving **0.959–0.972 weighted F1** across five lender benchmarks.
- **Develop OCR-free classification models** using **Donut/Swin**, fine-tuning **75M parameters** with PyTorch Lightning DDP across **20 L40S GPUs using BF16**; export models to ONNX after benchmarking **9 SageMaker instance types**.
- **Recover a production accuracy regression** by fixing the train-to-benchmark taxonomy mismatch that took F1 from **0.62 to 0.92**; establish that **64% of residual errors were valid predictions discarded by confidence thresholds**, leading to calibrated **per-class selective prediction across all 643 classes**.

CORE COMPETENCIES

- Machine Learning & Multimodal Deep Learning
- Computer Vision & OCR-Free AI
- NLP & Information Extraction
- LLM Engineering & Generative AI
- RAG & Enterprise Retrieval Systems
- Transformer Architecture & Model Fine-Tuning
- Distributed ML & Large-Scale Model Training
- MLOps & Production AI Engineering
- AWS SageMaker & Cloud AI Engineering
- Model Optimization, Inference & AI Reliability
- AI Solution Architecture & Technical Leadership

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** PyTorch | PyTorch Lightning | TensorFlow | Keras | Hugging Face Transformers | CNN | ResNet50 | BiLSTM | Swin Transformer | Vision Transformer (ViT) | Donut | Multimodal Fusion | Named Entity Recognition (NER) | LoRA | QLoRA | PEFT | Supervised Fine-Tuning (SFT) | Instruction Tuning | RLHF/DPO | Mixed Precision (FP16/BF16)
- **Generative AI, LLM & RAG:** Large Language Models (LLMs) | Generative AI | Retrieval-Augmented Generation (RAG) | FAISS | Dense Retrieval | Okapi BM25 | Hybrid Retrieval | Reciprocal Rank Fusion (RRF) | Cross-Encoder Reranking | Structured Outputs | JSON Schema | Llama 3 | Anthropic Claude API | Selective Prediction | Confidence Calibration
- **Computer Vision & Document AI:** Computer Vision | OCR/OCR-Free Processing | OpenCV | Tesseract | AWS Textract | TensorFlow Object Detection | Image Preprocessing | Document Classification | Information Extraction | Object Detection | Signature Verification | Document Deduplication
- **Distributed ML & Data Engineering:** Distributed Data Parallel (DDP) | Horovod | TorchDistributor | NCCL | Multi-GPU Training | Apache Spark | PySpark | Databricks
- **MLOps, Deployment & Model Serving:** AWS SageMaker | MLflow | Model Registry | Real-Time Model Endpoints | Auto Scaling | TensorFlow Serving | ONNX | Docker | Jenkins CI/CD | AWS CloudWatch | Model Optimization | Production Monitoring
- **Programming, Databases & Platforms:** Python | SQL | Git | Linux | Bash

PERSISTENT SYSTEMS | Senior ML Engineer | Client: ICE Data Services | Hyderabad | Aug 2021 – Nov 2023

- **Engineered the foundational data ingestion and OCR pipeline** supporting ICE's production document classification ecosystem, processing encrypted lender image sets, extracting text at scale with **Tesseract**, and consolidating **27+ source schemas** into a unified labelled corpus using **PySpark UDFs**.
- **Developed an OpenCV-based document preprocessing pipeline** incorporating thresholding, denoising, skew correction, and image normalization, improving OCR quality and downstream model performance across diverse lender inputs.
- **Built the first multimodal CNN + BiLSTM training pipeline** on Databricks, including the BiLSTM encoder-decoder responsible for generating sequence representations consumed by downstream text-processing branches.
- **Established the initial AWS SageMaker deployment and monitoring architecture**, together with **MLflow experiment tracking**, providing the foundation for the subsequent production MLOps and Jenkins-to-SageMaker release pipeline.
- Supported the evolution of the initial ML architecture into the **production-scale 643-class document classification ensemble**.

ZYCLYX | Data Scientist | Client: Al Rajhi Bank, Riyadh | Riyadh | Aug 2019 – Mar 2021

- **Designed and built the bank's document digitization platform as its first ML Engineer**, processing approximately **100K bilingual Arabic/English documents** through an end-to-end classification, detection, OCR, and entity-extraction pipeline.
- Developed a **ResNet50 document classifier covering ~65 document types**, achieving approximately **88% classification accuracy**, followed by TensorFlow Object Detection for field localization, region cropping, Tesseract OCR, and GRU-based NER for extracting **names, countries, identification numbers, and other key entities**.
- **Engineered a deep-learning signature verification model** to identify potentially forged signatures and strengthen document-level fraud detection.
- **Diagnosed the primary upstream source of OCR errors** by auditing scanning configurations across major bank branches; identified inconsistent scanner settings causing colour distortion, rotation, and skew, and established a standardized configuration subsequently mandated bank-wide through an official bank circular.
- **Integrated ML extraction services into the bank's BluePrism RPA workflow** with a human-in-the-loop validation layer, enabling page-level and field-level verification while feeding corrections back as training signals.
- **Productionized the platform on-premise using Flask and Gunicorn microservices**, ensuring sensitive documents remained within the bank's network.
- **Mentored 5+ engineers** as the ML team expanded, accelerating their capabilities across OCR, document processing, model development, and training workflow.

VISSINDIA | Python & Computer Vision Engineer | Hyderabad | Feb 2017 – Aug 2019

- **Developed real-time computer-vision solutions** for identity verification and multi-object tracking using **Haar Cascades, HOG, CNNs, SIFT, SURF, FLANN, TensorFlow Object Detection, and OpenCV**.
- Built image and video preprocessing pipelines to improve the quality and consistency of noisy visual inputs before model inference.
- Developed detection, feature-matching, and tracking workflow for real-time video analytics and identity-verification applications.

MAJOR PROJECTS

Rulebook-RAG | Retrieval-Grounded Document Classification | *Personal Project | 2026*

- Built a **36-class RAG classifier** without labelled training data or model retraining, using a dynamically indexed rulebook.
- Enabled **zero-retraining class expansion**, achieving **0.83 accuracy** on 5 newly added classes.
- Implemented **BM25 + dense retrieval + RRF**, improving accuracy by **5.2 pts**; cross-encoder reranking added another **3.9 pts**.
- Delivered **citation-grounded predictions** with live rule validation, achieving **0 invalid citations across 775 predictions**.

Document Deduplication Pipeline | ICE Data Services

- Built near-duplicate detection across a **7.1M-page mortgage corpus** using **Donut/Swin embeddings across 40 GPU nodes**.
- Implemented **two-tier FAISS cosine search** at page and document levels using mean-pooled embeddings.
- Optimized exact retrieval through **label/page-count grouping** and transitive **Union-Find clustering**.
- Identified **67% of pages and 69% of documents** as near-duplicates for large-scale corpus rationalization.

EDUCATION

- **B.Tech. - Computer Science & Engineering:** TKR Engineering College, JNTU, Hyderabad | **2016**